



Mark Scheme (Results)

January 2023

Pearson BTEC Nationals

In Applied Science (31617H/1P)

Unit 1: Principles and Applications of Science I

Physics

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Unit 1: Principles and Applications of Science I Physics

General marking guidance

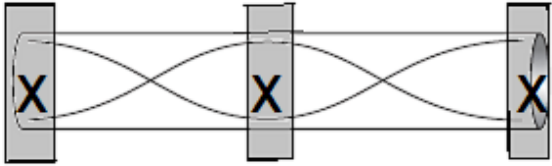
- All learners must receive the same treatment. Examiners must mark the first learner in exactly the same way as they mark the last.
- Mark grids should be applied positively. Learners must be rewarded for what they have shown they can do rather than be penalised for omissions.
- Examiners should mark according to the mark grid, not according to their perception of where the grade boundaries may lie.
- All marks on the mark grid should be used appropriately.
- All the marks on the mark grid are designed to be awarded. Examiners should always award full marks if deserved. Examiners should also be prepared to award zero marks, if the learner's response is not rewardable according to the mark grid.
- Where judgement is required, a mark grid will provide the principles by which marks will be awarded.
- When examiners are in doubt regarding the application of the mark grid to a learner's response, a senior examiner should be consulted.

Specific marking guidance

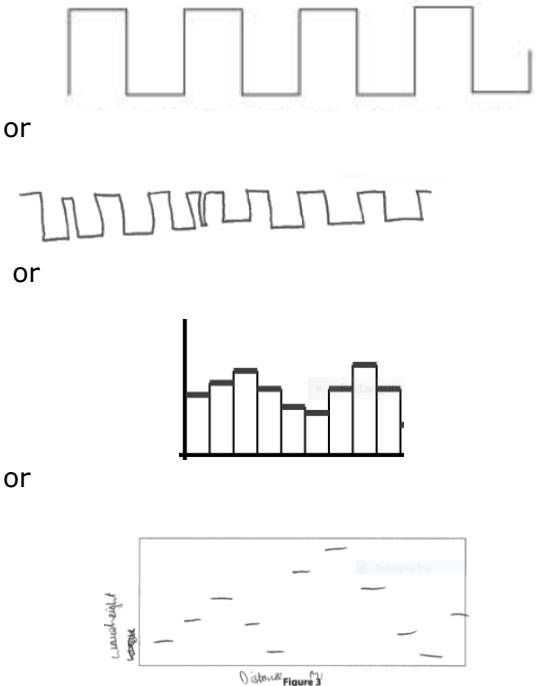
The mark grids have been designed to assess learners' work holistically.

Rows in the grids identify the assessment focus/outcome being targeted. When using a mark grid, the 'best fit' approach should be used.

- Examiners should first make a holistic judgement on which band most closely matches the learner's response and place it within that band. Learners will be placed in the band that best describes their answer.
- The mark awarded within the band will be decided based on the quality of the answer in response to the assessment focus/outcome and will be modified according to how securely all bullet points are displayed at that band.
- Marks will be awarded towards the top or bottom of that band depending on how they have evidenced each of the descriptor bullet points.

Question Number	Answer	Additional Guidance	Mark
1 (a)(i)	B - 10 cm		1
1 (a)(ii)	Any one position for X 	An X anywhere in any of the boxes. Judge by eye No mark is awarded if any X is drawn incorrectly	1
1 (b)	Substitution (1) $330 = 1370 \times \lambda$ Rearrangement (1) $(\lambda =) \frac{330}{1370}$ Evaluation (1) 0.24 (m)	Answer 0.24 m without working gains full marks Substitution and rearrangement in either order $\lambda = \frac{v}{f}$ Allow 0.2/0.241/ 2.4×10^{-1} 0.24 to any other power of 10 gains 2 marks 4.15 gains 2 marks 4.15 to any other power of 10 gains 1 mark $452100/4.5 \times 10^5$ with no working gains 1 mark	3

1 (c)	Any two from: (change) length (of pipe) (1) Close (one end of) the pipe (1) overblow/ blow hard(er) (1)	Ignore 'use different holes' and 'diameter' Change {pressure/ speed/velocity} of air	2
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Question Number	Answer	Additional Guidance	Mark
2 (a)	 or or or	Allow any correct written description Binary/series of zeros and ones Judge vertical lines by eye Diagrams with both analogue and digital signals must have digital signal labelled to gain the mark	1
2 (b) (i)	Any two from: Fibre optic cable (1) Radio-waves/ microwaves (1) Satellite (1)	Ignore copper/ wire cables Ignore 4G and 5G Ignore electromagnetic (em) waves	2
2(b) (ii)	multiplexing/ multiplex		1

Question Number	Answer	Additional Guidance	Mark
3 (a)	C is slower in glass / move towards the normal		1
3 (b)	Substitution (1) $1.52 = \frac{1}{\sin c}$ Rearrangement (1) $(\sin c =) \frac{1}{1.52}$ Evaluation of $\sin c$ (1) $(\sin c) = 0.658$ Evaluation of c (1) $41.1(^{\circ})$	Answer $41.1(^{\circ})$ without working gains full marks do not allow 41° without working Substitution and rearrangement in either order Allow $\sin c = \frac{1}{n}$ OR $(\sin 41 =) \frac{1}{1.52}$ Allow 0.657 Allow $\sin 41 = 0.656$ Allow answers rounding to 41.1° Allow 4 marks if evaluation in body of work is correct but transcribed to 41° in answer line Allow 3 marks for evaluation of 41.3/44.4	4
3 (c)	Award one mark for an identification point and two marks for further expansion points. Any of the following can be identification or expansion, depending on how the learner shapes their response. The critical angle for glass-air is smaller than the critical angle for glass-water (1) For reflection (to occur) angle of incidence is greater than the critical angle (1)	ORA may be implicit in other marking points	3

	For refraction (to occur) angle of incidence is less than the critical angle (1) In Figure 6a, 45°(angle of incidence) less than 62° (critical angle)(1) (In Figure 6a), water droplet increases the critical angle (1) In Figure 6b, 45° (angle of incidence) greater than 41°(critical angle) (1) In Figure 6b, total internal reflection/TIR occurs (1)	Figure 6a/glass-water interface must be identified Figure 6b/glass-air interface must be identified If no other mark awarded credit may be given for either of the following statements: Speed of light in air and water is different (1) Air and water have different (optical) densities	
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Question Number	Answer	Additional Guidance	Mark
4 (a)	C transverse with different frequencies		1
4 (b)	Award one mark for identification and one mark for the appropriate expansion. Identification (1) the idea of (base stations) have limited capacity of receiving (mobile phone signals) Expansion (1) too much call traffic/ more mobile telephones being used OR Identification (1) low power output /weaker signal Expansion (1) the idea of area coverage is small /limited range OR Identification (1) attenuation/reduced signal strength Expansion (1)		2

	buildings absorb/ reflect signals /cause interference		
4 (c)	Award one mark for identification and one mark for appropriate expansion. Identification Bluetooth® transmits radio waves/microwaves Expansion one from Not (easily) absorbed by walls (1) or (Radio-waves) diffract around objects (1)	Ignore frequency hopping Accept short wavelength (radio-waves)/high frequency (radio-waves) (1) Passes through walls/objects/obstacles	2

Question number	Indicative content
5	Answers will be credited according to the learner's demonstration of knowledge and understanding of the material, using the indicative content and levels descriptors below. The indicative content that follows is not prescriptive. Answers may cover some or all of the indicative content but learners should be rewarded for other relevant answers. Producing emission spectra - gases are heated - light is produced by heated gases - electrons are in (energy levels/shells) of atoms/molecules /particles (credit labelled diagrams) - when gases are heated electrons gain energy - electrons jump to a higher energy level /excitation - electrons (in an unstable state) return to lower energy level (accept de-excitation) - electrons lose energy as light (credit labelled diagrams) - no two elements have exactly the same energy levels/energy level changes - light emitted only has specific frequencies / $E = hf$ - light is passed through a grating - light is diffracted - constructive interference gives lines of specific frequencies Identification of elements - emission spectra show lines of different colour/wavelength/frequency - emission spectra of the gas is compared with (standard)emission spectra of elements - no two elements have lines of the same colour/wavelength/frequency

		• if some lines (in the emission spectra) of the gas match (exactly) lines (in the emission spectra) of an element • the element is present in the gas • wavelength or frequency of lines in gas and elements can be compared • matching lines can be looked for visually or by using a computer programme Ignore line thickness, dispersion by prisms, pH, flame tests, or any other chemical tests
Mark scheme (award up to 6 marks) refer to the guidance on the cover of this document for how to apply levels-based mark schemes*.		
Level	Mark	Descriptor
Level 0	0	No rewardable material.
Level 1	1–2	• Demonstrates adequate knowledge of scientific facts/concepts with generalised comments made • Generic statements may be presented rather than linkages being made so that lines of reasoning are unsupported or partially supported • The discussion shows some structure and coherence
Level 2	3–4	• Demonstrates good knowledge and understanding by selecting and applying some relevant scientific knowledge facts/concepts to provide the discussion being presented. • Lines of argument mostly supported through the application of relevant evidence • The discussion shows a structure which is mostly clear, coherent and logical
Level 3	5–6	• Demonstrates comprehensive knowledge and understanding by selecting and applying relevant knowledge of scientific facts/concepts to provide the discussion being presented. • Line(s) of argument consistently supported throughout by sustained application of relevant evidence • The discussion shows a well-developed structure which is clear, coherent and logical



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